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Study Protocol - Drug Checking Network

Study Rationale:

The ultimate goal of this research is to give Canadians data regarding drug consumption. Currently, there is very limited access to drug-checking across Canada for people who use drugs (PWUD). The lack of drug-checking services leads to people consuming substances that they do not intend to consume such as fentanyl and its analogs. This leads to accidental overdoses and deaths. In this study, drug samples will be obtained by participants at healthcare sites equipped with Section 56 approved drug-checking services. By providing these sites with drug-checking services, people will be more informed about the composition of their drugs and are hypothesized to discard samples more frequently and decrease their consumption more often. Drug-checking projects have been implemented across Canada, in Australia, Sweden, Slovenia, Hungary, Germany, Spain, France and Portugal (see references) with positive results in settings such as harm reduction centres, music festivals and clinics. Europe, where there have been various deaths caused by the consumption of PMA and PMMA, two synthetic drugs that are similar to MDMA but much more potent and toxic. Drug-checking has been implemented across the world to detect synthetic adulterants such as these.

Research Questions:

The first research question is: Can a real-time drug-checking network be implemented across multiple cities in Canada?
Our research hypothesis is that the knowledge of the content of a drug substance may influence the user to lower their consumption or discard their drug sample, thus reducing potential harm.

Outcomes:

The study outcomes include;

- Distribution of 10 Raman mapping systems to supervised consumption sites (SCS) and one in the chemistry laboratory at Western University.
- Implementation of a mass spectrometer for drug-checking at Western University – once a Section 56 Exemption is provided
- Implementation of an online platform that can integrate real-time drug-checking data across multiple cities in Canada for the public.
- Reporting of how drug-checking affects a participant's likelihood to lower their dose or discard their drug sample.

Protocol:

The participants will be recruited by harm reduction workers who will explain the process and indicate that consent is implied if a sample is provided for analysis. Any visitor of an SCS (above the age of 18) is eligible to participate in the project. Then, participants will be asked to provide

what the expected substance was (pre-drug-checking data input). They will also be asked if they would like to provide an anonymous client code (a code they make up) to allow them to access previous scans. They will be provided this code in writing along with a letter of information on the study. If they opt out, they can get immediate results but will not be able to access previous results specific to their sample (they will still be able to access the public study drug-checking network results). They will be asked if they would like a short scan (5 minutes) or a long scan (15 minutes). While participants are awaiting the results of the testing they will be asked demographic information which they may or may not choose to provide.

Sites:

Sites will normally be recruited as those already in agreement or considering using a device with the Scatr team. If necessary, our research team will engage in direct recruitment of new sites via email to the site administration using publicly available contact information.

The following organizations are partner sites under Collaborative Site Agreements with Western University. Each operates a Scatr device under its own Health Canada subsection 56(1) exemption and contributes data to the project database; sites whose agreement was executed in 2026 begin data collection once their exemption and device are in place.

Location	Name	Address	REB Approval (Local / UWO / Both)
1) London, ON	Regional HIV/AIDS Connection	446 York Street, London, ON (Carepoint supervised consumption site; organization offices 201-602 Queens Ave, London, ON N6B 1Y8)	UWO
2) Guelph, ON	Guelph Community Health Centre	176 Wyndham Street N. Guelph, ON. N1H 8N9	UWO
3) Niagara, ON	Positive Living Niagara	120 Queenston St, St. Catharines, ON L2R 2Z3	UWO
4) Kitchener, ON	Sanguen Health Centre	130 Victoria Street South, Kitchener, ON N2G 2B5 (fixed drug checking site and mobile unit)	UWO
5) Ottawa, ON	Ottawa Inner City Health	256 King Edward Avenue, Ottawa, ON (Trailer 2.0 supervised	UWO

		consumption site; organization offices 550 Old St Patrick Street, Ottawa, ON K1N 5N7)	
6) Ottawa, ON	Sandy Hill Community Health Centre	221 Nelson Street Ottawa, ON, K1N 1C7	UWO
7) Hamilton, ON	Hamilton Urban Core Community Health Centre	181 Main St W #121, Hamilton, ON L8L 4S1	UWO
8) New Westminster, BC	Lower Mainland Purpose Society for Youth and Families	40 Begbie Street New Westminster, BC V3M 3L9	UWO
9) London, ON	Western University/SCATR	1151 Richmond Street	UWO
10) Owen Sound, ON	County of Grey	595 9th Avenue East Owen Sound, Ontario, N4K 3E3	UWO
11) Brampton, ON	Moyo Health	7700 Hurontario St. #601, Brampton, Ontario L6Y 4M3	UWO
12) Extra Unit: London, On	Regional HIV/AIDS Connection	446 York Street, London, ON (Carepoint supervised consumption site; organization offices 201-602 Queens Ave, London, ON N6B 1Y8)	UWO
13) Moncton, NB	Ensemble Moncton	80 Weldon Street, Moncton, NB	UWO
14) Timmins, ON	Cochrane District Paramedic Service (mobile)	500 Algonquin Boulevard East, Timmins, ON	UWO
15) Renfrew, ON	Renfrew Paramedic Services (mobile)	450 O'Brien Road, Renfrew, ON	UWO
16) Peterborough, ON	Peterborough AIDS Resource Network (PARN)	60 Hunter Street East, 2nd Floor, Peterborough, ON	UWO
17) Chicoutimi, QC	Service de travail de rue de Chicoutimi	221 rue Tessier, Chicoutimi, QC G7H 1K4	UWO
18) Truro, NS	NHC Society	76 Esplanade, Truro, NS	UWO
19) Toronto, ON	Breakaway Community Services	21 Strickland Avenue, Toronto, ON	UWO

20) Fredericton, NB	AIDS New Brunswick	354 King Street, Fredericton, NB	UWO
21) Saint John, NB	Avenue B Harm Reduction Inc.	62 Waterloo Street, Saint John, NB	UWO
22) Edmonton, AB	Boyle Street Community Services	#201, 14065 Victoria Trail, Edmonton, AB	UWO
23) Newmarket, ON	LOFT Community Services (mobile drug checking, York Region)	Mobile unit parked at 102 Main Street South, Newmarket, ON; offices 26 Wellington Street East, 10th Floor, Toronto, ON M5E 1S2	UWO
24) Stratford, ON	Canadian Mental Health Association Huron Perth Addiction and Mental Health Services (agreement executed 2026)	380 Hibernia Street, Stratford, ON N5A 5W3	UWO
25) Gatineau, QC	BRAS Outaouais (agreement executed 2026)	109 Wright Street, Gatineau, QC J8X 2G7	UWO
26) Windsor, ON	Windsor-Essex County Health Unit (mobile drug checking service delivered with the Mobile Outreach Support Team (MOST) unit of CMHA Windsor-Essex; agreement in execution)	1005 Ouellette Avenue, Windsor, ON N9A 4J8 (mobile unit stored at CMHA Safe Beds, 354 Aylmer Avenue, Windsor, ON)	Both (University of Windsor REB application in preparation)

Methods:

A trained harm reduction worker will place the drug sample onto an aluminum Scatr chip, which is then inserted into the Scatr Raman mapping system. Only harm reduction workers trained to handle controlled substances will be permitted to handle the drug samples as well as the on-site spectrometers and Scatr software for data recording. Only sites holding a Health Canada subsection 56(1) exemption to conduct drug checking (supervised consumption services, harm reduction organizations, public health units and other healthcare organizations offering drug checking) will be permitted to participate in the study. The worker will subsequently activate the Raman spectroscopy system via Bluetooth through the online Scatr Dashboard (on a site laptop). The Raman data is processed on the Scatr Dashboard in three forms; The first is a 3D spatial map of the sample on the Scatr chip, with each point (pixel) representing a spot of the sample that has been scanned and classified by the machine learning

(ML) algorithm. The second is a table displaying a list of the substances that have been detected by the ML and their estimated percent composition ranges. The third form of analysis is the Similar Samples feature, where the sample will be statistically compared to all other street-obtained sample spectra and outputted. If a sample that has previously been scanned and associated with an overdose is statistically similar to the drug being scanned, an additional warning will appear, notifying the participant. The first two forms compare the street sample to the reference standard Scatr library, whereas the Similar Samples feature compares the street sample to other street samples that have been scanned at any of the participating SCSs. Clients will not receive guidance, advice, or recommendations on consumption based on results, as each individual's biochemical ability to process drugs are unpredictable, thus making recommendations risky. Participants who decline to take the LOI with them will be offered the 1-page project info sheet to increase likelihood they leave with some project information in hand.

To enhance site user comfort with drug checking as well as increase capacity to check drugs at sites, project sites and Scatr will be providing training to interested peers to become 'Drug Checking Peers'. Participating peers will be invited to complete a pre- and post-training survey as part of our research, to provide us with some descriptive information regarding this training.

Dissemination of Results:

Results will be shown to the participant via a laptop monitor handled by the trained harm reduction worker. If the participant makes the choice not to consume the sample. Once a Section 56 Exemption has been received by Western University they will have the option to surrender it for further testing, their sample will be sent to Western University for an additional Raman spectroscopy scan and mass spectroscopy testing (to provide quantitative data). Their SCS patient code will be included with the sample for the purpose of returning the results. The quantitative results will be available in the data portal within two weeks of sample submission in addition to Raman spectroscopy data acquired at the laboratory, using the testing code they were provided. In addition, a hard copy of the result will be returned to the SCS with the patient

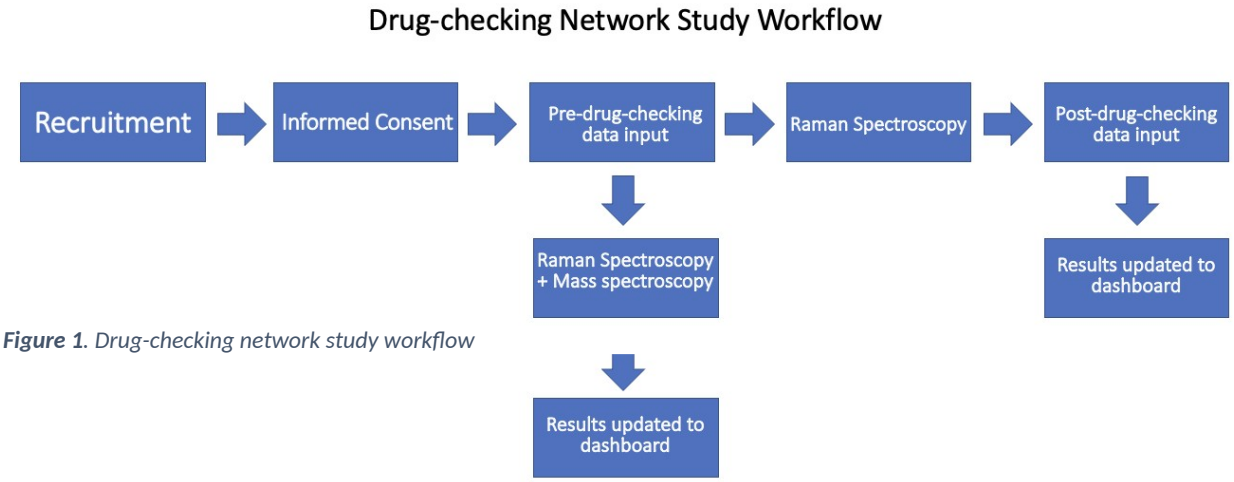


Figure 1*. Drug-checking network study workflow*

code indicated. As mass spectroscopy destroys samples, samples will not be returned to clients. A subset of the samples analysed at Western University is forwarded to Health Canada's Drug Analysis Service (DAS) for a further level of confirmatory analysis; DAS results are returned to the data portal in the same way as the Western laboratory results.

Data Collection:

The following is the proposed Data Collection Dashboard of the software running the instrument. Each bullet point corresponds to a field which will be filled by both the user under the supervision of the site worker.

- Staff member code
- Serial #
- Client Code
- Week
- Expected substance
- Description of substance

Demographic Questions

- What is your gender? _____

- What is your age? ___
- Do you live with chronic pain? (Y/N)
- Do you identify as Indigenous? (Y/N)
- Do you identify as Black, racialized or a person of colour? (Y/N)
- What is your preferred language? _____
- Do you identify as LGBTQ or another minority sexual orientation or gender identity? (Y/N)

Results

- Scan impact (ex. dispose of substance, surrender substance for further testing, use less substance, use later, no difference)
- Reported prior overdose with this sample? (Y/N)
- Overdose with today's use of the sample? (Y/N)
- Further notes on sample
- Upload data now?

Clients entering the study can choose to be assigned an anonymous client code (optional) in order to access their previously scanned samples. If the participant does not want to create a client code they can receive the on-site results, however will not be able to access the scan again or access the additional results from Western mass spectrometry if they chose to surrender their sample (once this option is available via Section 56 exemption).

In the **impact** of the scan section, the harm reduction worker will report how the result of the scan affected the participant's decision to consume (i.e. whether they chose to discard their sample, surrender their sample for further testing and destruction (once this option becomes available), use less of their sample than intended, or wait to use their sample later) by asking the participant verbally how the results will affect their consumption. A worker will ask the participant, "Will your drug-checking result lead you to lower your drug dose, discard the sample, surrender the sample for [further testing and] destruction, wait longer to use your drug, or other (text box available for other)? If the same sample has also been associated with a past overdose (self-reported by a participant), the harm reduction worker will record that for the purpose of warning participants that bring in samples with statistically similar samples for drug-checking. If a participant scans a sample and a warning system reports that a previously scanned sample which is significantly similar to their sample was associated with an overdose, it is hypothesized that they will choose to lower their dose or not consume at all. If the sample is used and results in an overdose in the SCS, this information will be reported in the overdose section of the record.

In the notes section, harm reduction workers can add any relevant additional notes about the drug sample i.e. estimated quantity (only information about the sample will be permitted here). The share notes button allows the harm reduction worker to choose to upload the results of the drug-checking data to the drug-checking network. This allows the harm reduction worker to

make any additions to the record, such as an overdose occurring, before the data is uploaded to the shared database. Only drug-checking results and study outcomes will be shared in the public drug-checking network; client codes will never be shared to the network. The network will include data such as which drugs, fillers, and adulterants are being detected in each city based on the results occurring from each site-specific login.

Drug Checking Peers

To improve uptake of drug checking and continuation of this study, additional funding has been received from the Health Canada SUAP stream to add 'Drug-Checking Peers' (DCPs). DCPs will be people with lived or living experience of substance use who are trained on the Scatr device by Western and Scatr staff, who will then promote and assist use of the device amongst their peers. DCPs will be supervised and supported by site operators. DCPs will therefore follow usual adherence to the rules, regulations and supervision of their respective sites, with the formalization of their role depending on the volunteer or staffing requirements of each site. The Western Team will be responsible for intermittent follow-up with the site leadership and Drug Checking Peers to ensure continued project fidelity as needed and in accordance with all previously established data collection protocols. Pre- and post-training evaluation will be conducted via survey to constantly iterate and improve the training to optimize DCP knowledge and empowerment. Across the project sites and 4 additional years of funding, we anticipate a maximum sample size of 350 DCPs.

The Drug-Checking Peer (DCP) training will be delivered by a member of the Western Research Team either **in-person, on-site**; or, **virtually, on-site, via Western Corporate Zoom**. This optionality was recommended and approved by our Health Canada Senior Project Advisor with the aim of providing more scheduling flexibility and decreased travel-related risks during the winter months, in addition to the concomitant potential cost-savings.

If the Drug-Checking Peer (DCP) training is to be conducted with the Western Research Team member **in-person, on-site**, the following protocol will apply:

- The Letter of Information (LOI) will be provided to prospective participants in advance of the training date, concomitant with the recruitment of potential participants. The Primary Site Contact at each location will receive LOIs via email from the Western University team, print them, and distribute them in person to prospective participants. Participants will be instructed to bring their LOI with them to the training session. Extra paper copies will be made available by the Primary Site Contact on the day of training, should anyone forget to bring theirs. This advance distribution allows participants time to review the information, consider their participation, and prepare any questions before the training session.

- On the day of training, prior to the training session beginning, the Western Research Team member will read the LOI document aloud to the group. Following the reading, the trainer will ask if anyone has questions, they are comfortable asking in front of the

group. The trainer will then ask if anyone has questions, they would like to ask in private. For private questions a breakout room will be made available, allowing the participant to speak with the trainer privately. Once finished, the participant and Western Research Team member can rejoin the main group. Consent for survey participation is obtained through in-person, wet-ink, co-signing of the participant and Western Research Team member, as observer.

- Participation with the surveys is 100% voluntary. Participants may choose not to answer any or all questions, or stop participating at any time, without any consequences to their participation in the DCP training. Participants who decline to complete the surveys may still attend the full training and receive their certificate of completion.

- The pre- and post-training surveys will be conducted in person, on-site, on paper, on the day of training. The invitation to participate will be made by the Western Team training member, to all subjects who are there to participate in that day's training. The inclusion criteria for the surveys will only require that they attended all 3 hours of training that day. Inclusion criteria for the training itself will be determined by participants eligibility as a PWLLE within their respective communities and most will be recommended by site operators as currently engaged members of the community. The analysis will consist of simple quantitative descriptive techniques and ANOVA.

If the Drug-Checking Peer (DCP) training is to be conducted with the Western Research Team member **virtually, on-site, via Western Corporate Zoom**, the following protocol will apply:

- The Letter of Information (LOI) will be provided to prospective participants in advance of the training date, concomitant with the recruitment of potential participants. The Primary Site Contact at each location will receive LOIs via email from the Western University team, print them, and distribute them in person to prospective participants. Participants will be instructed to bring their LOI with them to the training session. Extra paper copies will be made available by the Primary Site Contact on the day of training, should anyone forget to bring theirs. This advance distribution allows participants time to review the information, consider their participation, and prepare any questions before the training session.

- For virtual trainings, a designated Primary Site Contact at each location will facilitate on-site logistics, including managing audio-visual equipment, distributing materials, and assisting participants. The site will be equipped with AV equipment (TV/projector and audio system) to allow participants to view and hear the presentation, as well as a webcam or similar device to enable bi-directional visual communication between the trainer and participants. A separate electronic device will also be available to facilitate a private Zoom breakout room for participants who wish to ask questions privately.

- On the day of training, prior to the training session beginning, the Western Research Team member will read the LOI document aloud to the group. Following the reading,

they will ask if anyone has questions they are comfortable asking in front of the group. They will then ask if anyone has questions they would like to ask in private. For private questions during virtual trainings, a separate Zoom breakout room will be made available with a separate electronic device, allowing the participant to speak with the Western Research Team member privately. Once finished, the participant and Western Research team member can rejoin the main groups. Consent for survey participation is obtained through in-person, wet-ink, signing of the participant. Because the Western Research Team member will be present only virtually during this process, they will be unable to co-sign in real-time. However, at the recommendation of the projects Ethics team, the participant-signed hard copy LOIC's can be mailed to the Western Research Team member after the training session, for their final co-signing. Then an electronic copy will be emailed back to participants within 3 weeks' time. Therefor the Virtual Trainer-specific LOIC will now include the option to provide an email address.

- The pre- and post-training surveys will still be conducted in person, on-site, on paper, on the day of training. They will be sent in advance to the Primary Site Contact, to be printed out and made available for the day of. The invitation to participate will be made by the Western Team training member, to all subjects who are there to participate in that day's training. The inclusion criteria for the surveys will only require that they attended all 3 hours of training that day. Inclusion criteria for the training itself will be determined by participants eligibility as a PWLLE within their respective communities and most will be recommended by site operators as currently engaged members of the community. At the conclusion of the training session, participants will be given their Certificate of Completion (which have a blank name field, for participants to fill out only if they wish). Then, participants will be asked by the Western Research Team member to place their surveys (a single stapled packet) and signed LOIC form in a dropbox at the front of the room as they exit. The contents of the dropbox will subsequently be mailed to the Western Research Team. Upon receipt, the Western Research Team member will complete the process by co-signing as observer and a scanned electronic copy will be emailed back to participants for their records, within 3-weeks' time.

Important Notes

Training Access

Participation in the research (completing surveys) is voluntary and separate from attendance at the DCP training. All individuals who sign up for the training may attend the full 3-hour session and receive their certificate of completion, regardless of whether they choose to participate in the research component.

Private Question Accommodation

For virtual trainings, a separate Zoom breakout room with a separate electronic device is made available for participants who wish to ask questions privately. This ensures all participants have the opportunity to have their questions answered in a comfortable setting.

Virtual Training Equity

Virtual training delivery expands access to DCP training across geographically dispersed sites while maintaining the same rigorous data collection standards as in-person training. The Primary Site Contact role ensures that on-site support is available for all participants during virtual sessions.